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MINIMAL MODEL FOR THERMODYNAMICS OF NONEQUILIBRIUM PHASE TRANSITIONS

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- **Nonequilibrium phase transitions** are most commonly discussed in the framework of nonlinear dynamics and bifurcation theory.
- **Stochastic thermodynamics** provides a systematic treatment of nonequilibrium thermodynamics $\Rightarrow$ **thermodynamics of nonequilibrium phase transitions**.
- Stochastic thermodynamics applied to noninteracting systems or system made of few interacting degrees of freedom.
- **Practical motivation** to explore the thermodynamics of nonequilibrium phase transitions in large ensembles of interacting degrees of freedom:
 - Energy-conversion of nanomachines
 - Highly efficient but low power output
 - Phase transitions affect performance of ensembles of machines?
 - Role of the number of nanomachines?
- **Synchronization** as a promising candidate for a beneficial performance of the ensemble.

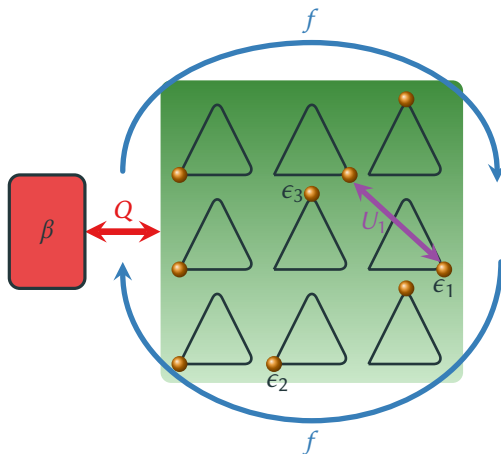
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- **Synchronization** as a promising candidate for a beneficial performance of the ensemble.

$\Rightarrow$ Search for a (feasible) thermodynamic model exhibiting synchronization.

[Herpich, Thingna, & Esposito arXiv:1709.06045]

OPEN THREE-STATE UNIT NETWORK

- **Network** of N all-to-all coupled three-state units [PRL **96**, 145701 (2006)] units.
- Open system in contact with **heat bath**.
- **Interaction energy** $U(\mathbf{N}) = \frac{u}{2N} \sum_{k=1}^3 N_k^2 + U_0$.
- Subjected to **torque** f acting as a global bias.



- **Markovian master equation** with Arrhenius rates satisfying **local detailed balance**

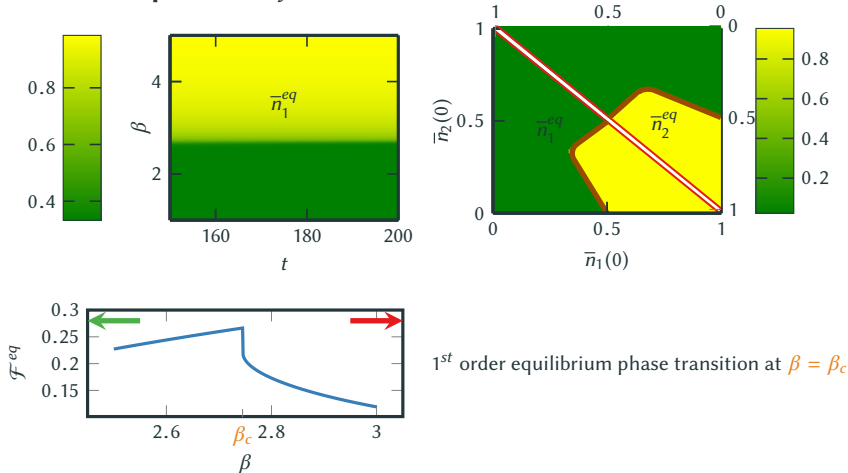
$$\dot{P}_{\mathbf{N}} = \sum_{\mathbf{N}'} W_{\mathbf{N}, \mathbf{N}'} P_{\mathbf{N}'}, \quad \ln \frac{W_{\mathbf{N}\mathbf{N}'}}{W_{\mathbf{N}'\mathbf{N}}} = -\beta [\Delta F^{eq}(\mathbf{N}, \mathbf{N}') - \sigma(\mathbf{N}, \mathbf{N}')f], \quad F^{eq}(\mathbf{N}) = E(\mathbf{N}) - \beta^{-1} \overbrace{\ln \frac{N!}{\prod_i N_i!}}^{S^{int}(\mathbf{N})}$$

- Mean-field dynamics given by a nonlinear master equation

$$\dot{\bar{n}}_i(t) \equiv \langle \dot{n}_i(t) \rangle = \sum_j k_{ij}(\bar{n}_i, \bar{n}_j) \bar{n}_j(t), \quad \ln \frac{k_{ij}}{k_{ji}} = -\beta [\epsilon_i - \epsilon_j + u(\bar{n}_i - \bar{n}_j) - \sigma(i, j)f].$$

- For $\epsilon_j = \epsilon$, the nonlinear equation has a **symmetric fixed point** $\dot{\bar{n}}_i = \sum_j k_{ij} \bar{n}_j \big|_{\bar{n}_j=1/3} = 0$.

Equilibrium $f = 0$ and attractive interactions $u < 0$



- Three distinct fixed points in the low-temperature regime ($\beta \gg \beta_c$)
- Symmetric solution in the high-temperature regime ($\beta \ll \beta_c$)

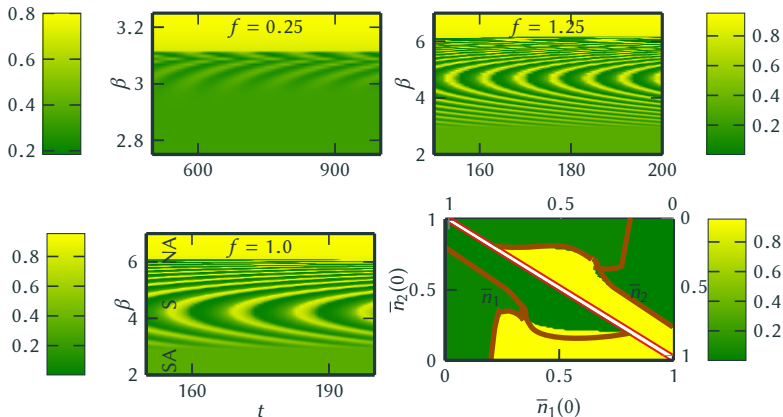
Energy-driven

Entropy-driven

- Linear stability analysis indicates a Hopf-Bifurcation at $\beta u = -3$

$$\lambda_{\pm} = -\Gamma(\beta u + 3) \cosh\left(\frac{\beta f}{2}\right) \pm i\sqrt{3}\Gamma \sinh\left(\frac{\beta f}{2}\right).$$

Nonequilibrium $f \neq 0$ and attractive interactions $u \equiv -1$



$\Rightarrow$ Two nonequilibrium phase transitions at $\beta = \beta_{c1,2}$.

- Low- and high temperature regime exhibits same phenomenology in and out-of-equilibrium.
- If sufficiently far away from equilibrium ($f > f_c$), stable oscillations emerge at intermediate temperatures ($\beta_{c1} < \beta \leq \beta_{c2}$).

Stochastic Dynamics

- The microscopic/mesoscopic dynamics is ruled by an irreducible and Markovian master equation $\dot{P}_{\mathbf{N}} = \sum_{\mathbf{N}'} W_{\mathbf{N},\mathbf{N}'} P_{\mathbf{N}'}$ and exhibits a unique steady state (PF theorem).

Mean-Field Dynamics

- The macroscopic dynamics is governed by a nonlinear master equation $\dot{\bar{n}}_i = \sum_j k_{ij} \bar{n}_j$ and displays stable oscillations ($\beta_{c1} \leq \beta < \beta_{c2}$) and multistability ($\beta \geq \beta_{c2}$).

⇒ **Non-commutation of the thermodynamic- and the stationary limit.**

$$\lim_{t \rightarrow \infty} \lim_{N \rightarrow \infty} \mathbf{P}(t) \neq \lim_{N \rightarrow \infty} \underbrace{\lim_{t \rightarrow \infty} \mathbf{P}(t)}_{\mathbf{P}^s}, \quad \text{if } \beta \geq \beta_{c1}.$$

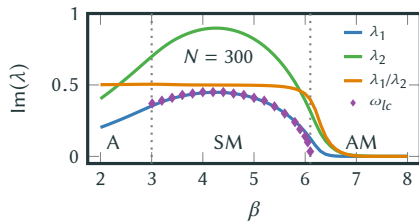
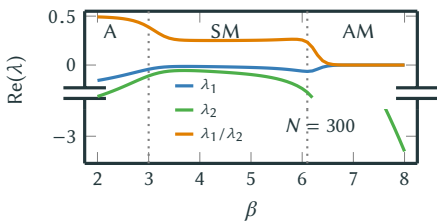
- Apparent paradox since mean field represents the exact solution in the $N \rightarrow \infty$ limit.
- Idea:** Infinite-time solutions of mean-field exhibited also by finite systems at long but *finite* timescales → **Metastability?**

- Formal solution of the finite-size master equation reads

$$\mathbf{P}(t) = e^{\mathbf{W}t} \mathbf{P}(0) = \sum_{i, i^*} \underbrace{\Phi_i^L \mathbf{P}(0)}_{\equiv c_j} e^{\lambda_i t} \Phi_i^R + c_{i^*} e^{\lambda_{i^*} t} \Phi_{i^*}^R,$$

- The PF theorem prescribes the existence of a non-degenerate $\lambda_0 \equiv 0$, while $\text{Re}(\lambda_{i \neq 0}) < 0$.
- Relevant timescales are specified by the slowest decaying modes.

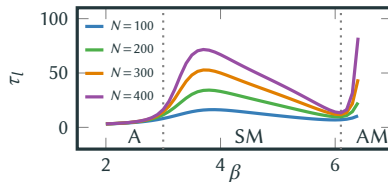
$$\text{Relaxation } \tau_r \sim -\frac{1}{\text{Re}(\lambda_1)}, \quad \text{Metastable } \tau_m \sim -\frac{1}{\text{Re}(\lambda_2)}, \quad \text{Lifetime } \tau_l \equiv \tau_r - \tau_m \sim \frac{\text{Re}(\lambda_1) - \text{Re}(\lambda_2)}{\text{Re}(\lambda_1) \text{Re}(\lambda_2)},$$



- Real-part gap between the two most dominant non-null modes is opening at both critical points $\rightarrow$ time-scale separation between the mode associated with λ_1 and $\lambda_{i>1}$.
- Imaginary part of the first eigenvalue is in very good agreement with the limit cycle frequency and vanishes as the second critical point is approached.

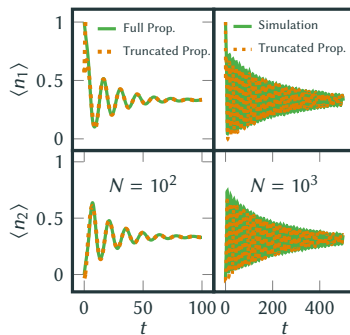
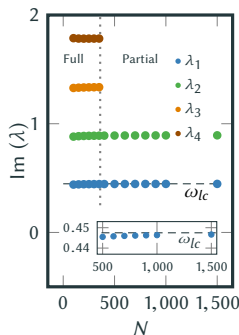
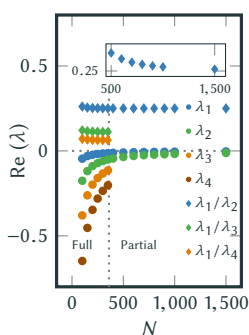
CONVERGENCE TO MEAN FIELD VIA METASTABILITY I

- The metastable states should become more long-living as N increases to ensure smooth convergence to mean field.

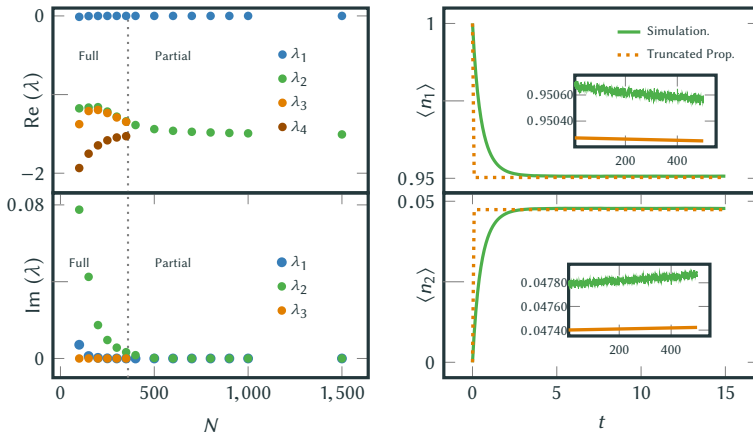


⇒ Study spectrum as a function of system size N

$\beta = 4.0$ (SM phase)

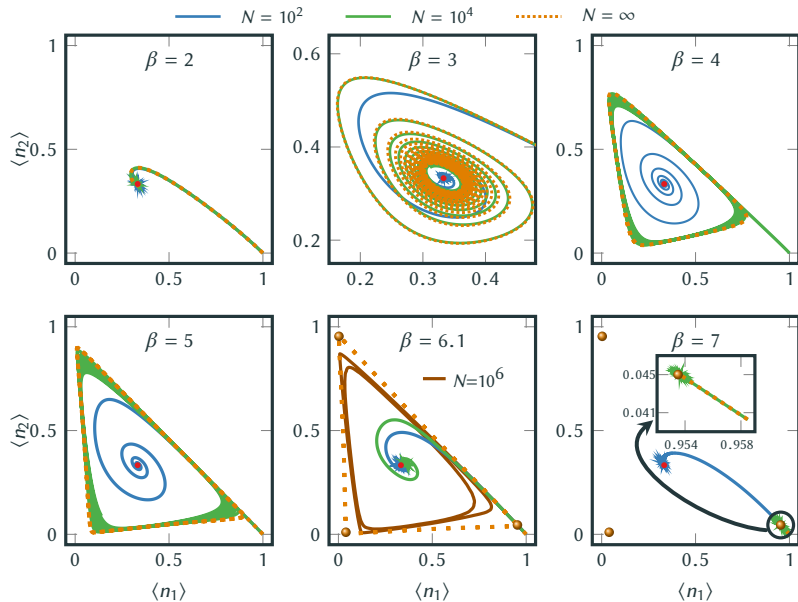


$\beta = 7.0$ (AM phase)



- Relaxation dynamics can be characterized only by the modes constituting the metastable state, *i.e.*

$$\mathbf{P}(t) \stackrel{t \gg \tau_m}{\approx} c_0 \Phi_0^R + e^{\lambda_1 t} c_1 \Phi_1^R + e^{\lambda_1^* t} c_1^* \Phi_1^{R*}$$

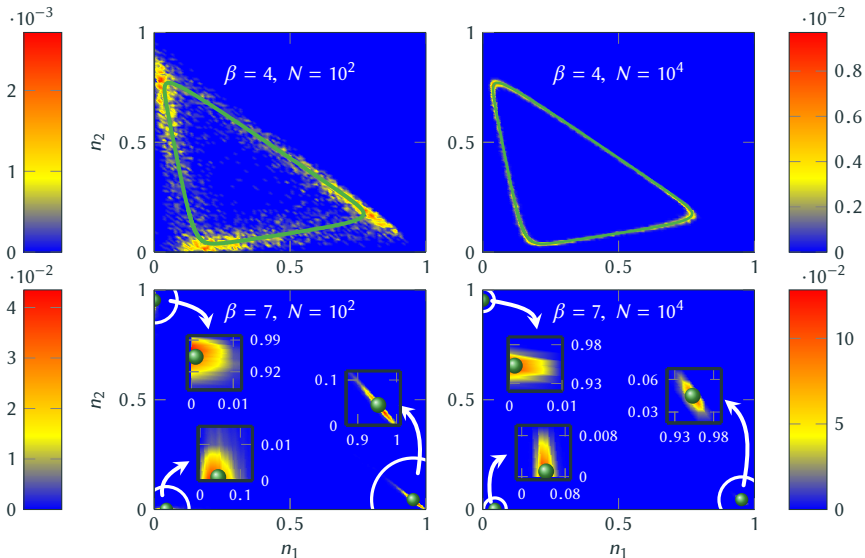


$\Rightarrow$ Large system dynamics indeed agrees with the mean-field solution at metastable times.

ROLE OF INITIAL CONDITION FOR METASTABILITY I

- Recall that $\mathbf{P}(t) \stackrel{t \gg \tau_m}{\approx} c_0 \Phi_0^R + e^{\lambda_1 t} c_1 \Phi_1^R + e^{\lambda_1^* t} c_1^* \Phi_1^{R*}$ with $c_{1,1^*} = \Phi_{1,1^*}^L \cdot P(0)$.
- Metastability can be removed from the dynamics for $P(0) = b_0 \Phi_0^{R*} + \sum_{i \neq 1, 1^*} b_i \Phi_i^{R*}$.

Is metastability a generic property of our system or merely an artifact of special initial conditions?



$$N = 10^2$$

$$\beta = 4$$

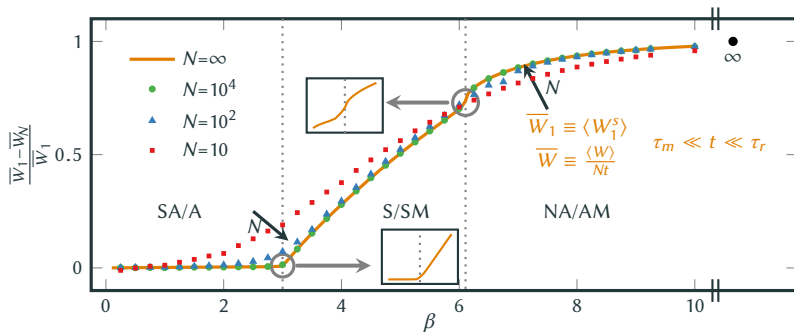
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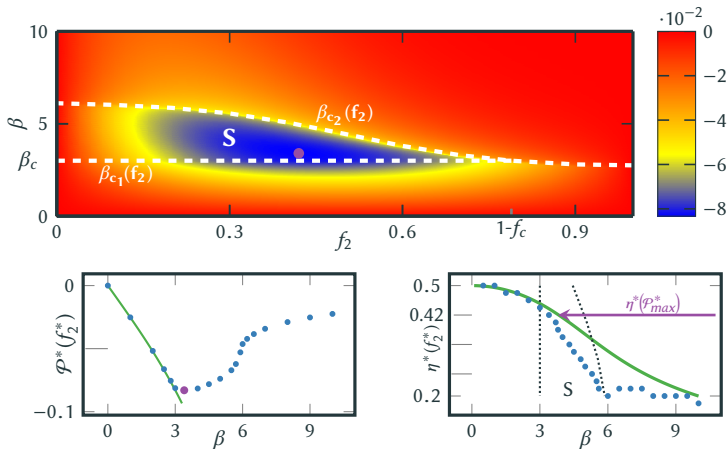
- **First Law:** $d_t \langle E \rangle = \langle \dot{Q} \rangle + \langle \dot{W} \rangle$
- With this setup, work needs to be done on average to maintain the system, $\langle \dot{W} \rangle > 0$, which is on average dissipated as heat into the bath, $\langle \dot{Q} \rangle < 0$.



- **(Signatures of) Nonequilibrium phase transitions** at the critical points (for large but finite N). [“First-order phase transition” at β_{c1} and “second-order phase transition” at β_{c2}].
- **Reduction** of the dissipated work for **finite interactions** $u < 0$.
- Dissipated work is **further reduced** for **smaller** networks in the two high-temperature regimes.

EFFICIENCY AT MAXIMUM POWER

- **Work-to-work conversion** $f \equiv f_1 - f_2$, with input work $\mathcal{W}_1(f_1)$ and output work $\mathcal{W}_2(f_2)$.
- **Maximization of the output power** $\mathcal{P} \equiv \dot{\mathcal{W}}_2$ with respect to output force $\partial_{f_2} \mathcal{P}|_{f_1=1} = 0 \Rightarrow f_2^*$.
- Efficiency of work-to-work conversion at maximum power $\eta^* \equiv 1 - \frac{f}{f_1}|_{f_2=f_2^*}$



- Power output is maximized when the network is **synchronized**, *i.e.* sufficiently far driven out-of-equilibrium $f \geq f_c$.
- The **efficiency at maximum power**, $\eta^*(f_2^*)$, is surprisingly **close** to the universal value, $\eta^* = 1/2$, for a tightly-coupled (single net-current) system responding linearly.

- Minimal thermodynamic model exhibiting synchronization.
- Model with two non-equilibrium and one equilibrium phase transition.
- Resolving the apparent paradox of a finite irreducible Markov process and mean-field dynamics displaying stable oscillations and multistability via notions of metastability.
- Reduction of the dissipated work for finite interactions. For high and intermediate temperatures, smaller networks are favorable.
- When operating as work-to-work converter, the maximum output is achieved for a synchronous operating mode. The associated efficiency of this far-from-equilibrium scenario is close to linear-response value.

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